CSA04-Operating Systems

TASK 8 – Evaluate CPU States and Protection

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1. User Mode and Kernel Mode

The hospital monitoring system uses two CPU operating modes to protect the system from unauthorized operations.

User Mode:

- Used to execute normal application programs.
- Applications have limited access to hardware and memory.
- They cannot directly execute privileged instructions.

Kernel Mode:

- Used by the operating system.
- Has full access to hardware and system memory.
- Performs important operations such as interrupt handling and memory management.

Privileged Instructions:

- Special instructions that can only execute in kernel mode.
- Examples include disabling interrupts and changing memory protection settings.

Mode Bit:

- A hardware bit that identifies the current CPU mode.
- It helps distinguish user mode from kernel mode.

2. Interrupts, Traps and Timer Interrupts

1. Interrupt: A signal from hardware that requests CPU attention, such as a sensor alert.
2. Trap: A software-generated event caused by a system call or an exception, such as division by zero.
3. Timer Interrupt: Generated periodically by a hardware timer. It allows the OS to regain CPU control and schedule other processes.

These mechanisms help the hospital system respond to emergencies and manage CPU time.

3. What Happens if Any Program Can Disable Interrupts?

If any application could disable interrupts:

- Emergency sensor alerts might be delayed or missed.
- Patient monitoring could stop responding.
- Other important processes might not receive CPU time.
- The operating system could lose control of system scheduling.
- It could create serious risks to patient safety.

Therefore, only the operating system should be allowed to disable interrupts through privileged instructions.

4. How Hardware Protects Memory

- Each process receives its own protected memory space.
- The Memory Management Unit (MMU) checks every memory access.
- Page tables define which memory locations a process can access.
- Unauthorized access generates a protection fault.
- The OS handles the fault and prevents the process from accessing another program's memory.

This protects patient records and monitoring applications from unauthorized access.

5. Alternative Concept and Complexity Analysis – Polling vs Interrupts

Polling: The CPU continuously checks the sensor at regular intervals to identify abnormal readings.

Interrupts: The sensor sends an interrupt signal to the CPU whenever an important event occurs.

Comparison

Factor	Polling	Interrupts
Working	CPU repeatedly checks sensors	Sensor alerts CPU
Response time	Depends on polling interval	Usually very fast
CPU usage	Higher	Lower when idle
Power consumption	Higher	Generally lower
Efficiency	Less efficient	More efficient

Worst-case Delay

Assume the hospital monitor checks the sensor every 100 milliseconds.

- Polling: An abnormal reading may take almost 100 ms to be detected.

- Interrupts: Detection depends on interrupt latency, which is generally much shorter than the polling interval.

$$Polling\ delay_{max} \sim T$$

Where T is the polling interval.

$$\text{Interrupt delay} \sim L$$

Where L is the interrupt latency.

CPU Time and Power Wasted by Polling

- The CPU repeatedly checks sensors even when there is no abnormal reading.
- This wastes processing time.
- Continuous checking increases power consumption.
- It reduces CPU availability for other hospital applications.

Evaluation: Interrupt-based monitoring is more suitable for hospital systems because it provides faster responses, reduces unnecessary CPU activity and improves power efficiency. However, interrupts must also be designed with appropriate priority and bounded response times.

6. Conclusion

1. User mode and kernel mode provide hardware protection.
2. Privileged instructions prevent unauthorized hardware operations.
3. Interrupts help the system respond quickly to emergency events.
4. MMU and memory protection prevent unauthorized memory access.
5. Interrupt-based monitoring is more efficient and reliable than continuous polling for critical hospital monitoring systems.

Result: CPU modes, interrupts and memory protection mechanisms help maintain the security, reliability and safety of hospital monitoring systems.